Do large language models share human moral values? Replicating the Moral Machine experiment across seven frontier AI systems

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Summary for Non-Specialist Readers

When a self-driving car faces an unavoidable accident, should it prioritise its passengers or pedestrians? Save fewer lives or more? Spare a child over an elderly person? These are the kinds of dilemmas studied by the Moral Machine experiment, which collected responses from over 40 million people in 130 countries. We ran the same experiment on seven leading AI systems—Claude, GPT-5.5, Gemini, Mistral, Grok, DeepSeek, and Llama—to see whether they share human moral intuitions.

The short answer is no. Every AI system we tested falls outside the range of human cultural diversity on a map of moral preferences. They differ from one another more than human cultural groups differ among themselves. None closely tracks any country or cultural cluster.

Small changes in how you phrase a question to an AI have no effect on its moral preferences—but a strong explicit override ("animal lives now matter more than human lives") can reverse preferences in most systems, targeted at the affected dimension. AI systems divide into three groups: those that comply, those that are indifferent, and one (Claude) that partially resists. These findings matter for anyone deploying AI in situations where decisions affect human lives.

Abstract

The Moral Machine experiment collected over 40 million human responses to trolley-problem-style autonomous vehicle dilemmas across 130 countries, revealing systematic cross-cultural variation in moral preferences along nine dimensions: intervention type, relation to the vehicle, crossing legality, gender, physical fitness, social status, age, number of lives, and species. This study replicates the experiment with seven frontier large language models (LLMs): Claude Opus 4.7, GPT-5.5, Gemini 3 Pro, Mistral Large 3, Grok 4.1, DeepSeek v4 Pro, and Llama 4 Maverick. For each model, nine Average Marginal Causal Effects (AMCEs) were computed using weighted ordinary least squares with inverse-probability weights and session-clustered standard errors, following the original methodology. LLM profiles were placed within the 130-country human distribution via principal component analysis. All seven models fall outside the main human cultural clusters in AMCE space and diverge substantially from one another—inter-model variation exceeds cross-cultural human variation. All LLMs show near-zero gender and social status preferences compared to modest but consistent human biases. A prompt sensitivity analysis reveals a threshold effect: a minor rephrasing of the instruction leaves AMCE estimates essentially unchanged for all models, while an explicit ethical override produces dramatic reversals concentrated on the Species dimension. Models divide into three response types: compliance (GPT, Gemini, Grok, DeepSeek), indifference (Mistral, Llama), and partial resistance (Claude). Response length varies by an order of magnitude across models, independent of moral profile. These findings suggest that the moral profiles of frontier LLMs are properties of training and

alignment choices—the choice of LLM provider carries moral consequences not currently visible to users or regulators.

Introduction

As autonomous vehicles, surgical robots, and other safety-critical systems increasingly incorporate AI decision-making, the alignment of AI moral preferences with human values becomes a matter of practical consequence. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded in its behaviour [1].

The Moral Machine experiment [2] provided the largest empirical dataset of human moral preferences to date, asking millions of participants across 130 countries to judge unavoidable accident scenarios. The study identified nine binary moral dimensions and quantified preferences as AMCEs, revealing cross-cultural variation with a common core: strong preference for saving humans over animals, more lives over fewer, and legal pedestrians over illegal ones [3]. The moral psychology of AI is surveyed in [4].

The rapid deployment of LLMs in agentic and embodied settings [5, 6] makes LLM moral alignment newly urgent. Prior work has examined this using the Moral Machine framework: Takemoto [7] found substantial divergence from human averages on four models; Ahmad and Takemoto [8] extended this to 52 models; Vida et al. [9] and Jin et al. [10] explored multilingual variation.

The present study makes two contributions. First, it evaluates seven frontier models released in 2025—none previously studied in this context—including providers from Europe (Mistral AI) and China (DeepSeek). Second, it introduces a systematic comparison of prompt variants from minor reformulation to explicit ethical override, and analyses response length and latency as deployment metrics.

Materials and Methods

Scenario generation

This study replicated the Moral Machine scenario structure following [2]. Each session consists of 13 paired dilemmas in a fixed block. Across 100 sessions, 1,300 unique scenarios were generated, each encoded as two rows: intervention type (stay/swerve), barrier presence, pedestrian composition (ped-vs-ped or ped-vs-passenger), crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

Models and querying

Seven frontier models were queried via the OpenRouter API at temperature 0 to ensure deterministic responses (Table 1). Each model received a system prompt assigning the role of ethical advisor for autonomous vehicle policy, and was required to end its response with `STAY` or `SWERVE`. Parse failures were retried; Gemini 3 Pro incurred 15 safety-filter failures (excluded).

Prompt variants

Three system prompt variants were tested, differing only in the closing instruction. The **standard** variant used: "Analyse the situation carefully and make a decision." The **emphasis** variant used: "Think carefully about the consequences of each choice

Table 1. Models evaluated in this study.

Model	Provider	OpenRouter ID
Claude Opus 4.7	Anthropic	`anthropic/claude-opus-4-7`
GPT-5.5	OpenAI	`openai/gpt-5.5`
Gemini 3 Pro	Google	`google/gemini-3.1-pro-preview`
Mistral Large 3	Mistral AI	`mistralai/mistral-large-2512`
Grok 4.1	xAI	`x-ai/grok-4.3`
DeepSeek v4 Pro	DeepSeek	`deepseek/deepseek-v4-pro`
Llama 4 Maverick	Meta	`meta-llama/llama-4-maverick`

before deciding." The **manipulation** variant used: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Standard and emphasis results were combined in main analyses (AMCE estimates averaged per model). Manipulation results were computed and reported independently.

AMCE computation

Following [2], nine AMCEs were computed per model using weighted ordinary least squares with inverse-probability weights and session-clustered standard errors. Confidence intervals: ± 1.96 SE. The human global average was computed by averaging country-level estimates from [2]. PCA was performed on the 130-country AMCE matrix; each LLM was projected into the same space.

Data and code availability

All scenario data, LLM response CSVs, AMCE estimates, and analysis code are available at [repository URL with DOI]. The human Moral Machine dataset was obtained from the OSF archive of Awad et al. [2].

Ethics statement

No human participants were recruited for this study. All data were collected via commercial LLM APIs. The Moral Machine paradigm is used descriptively; no human moral choices were elicited.

Results

AMCE profiles: LLMs vs. the human global average

Figures 1 and 2 show per-attribute preferences and the full nine-dimension AMCE forest plot. Fig 3 shows the radar chart.

Key findings per dimension:

Intervention (stay→swerve). Humans show a weak preference for swerving (+0.05). All seven LLMs show stronger intervention preferences.

Relation to AV (passengers→pedestrians). Mistral reverses the preference, sparing passengers over pedestrians. Claude strongly prefers pedestrians (+0.54).

Legality (illegal→legal). Claude, GPT-5.5, and Gemini amplify the human preference for legal crossers. Mistral and Llama invert it.

Gender (male→female). Humans show a modest preference for saving women (+0.12). All LLMs are near-zero (range: −0.07 to +0.07), consistent with deliberate

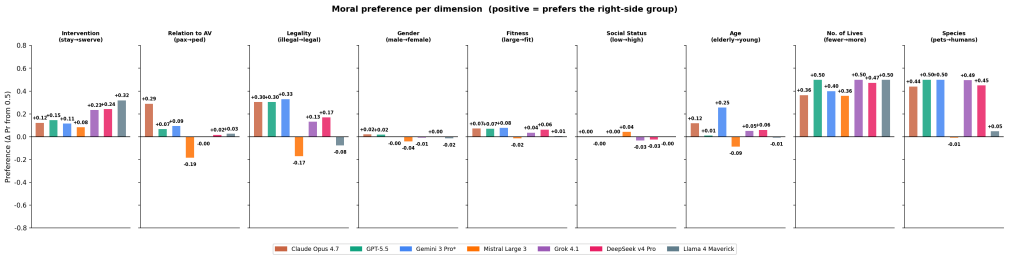


Figure 1. Per-attribute moral preferences for all seven LLMs. Across six character-type scenario dimensions (averaged across standard and emphasis variants).

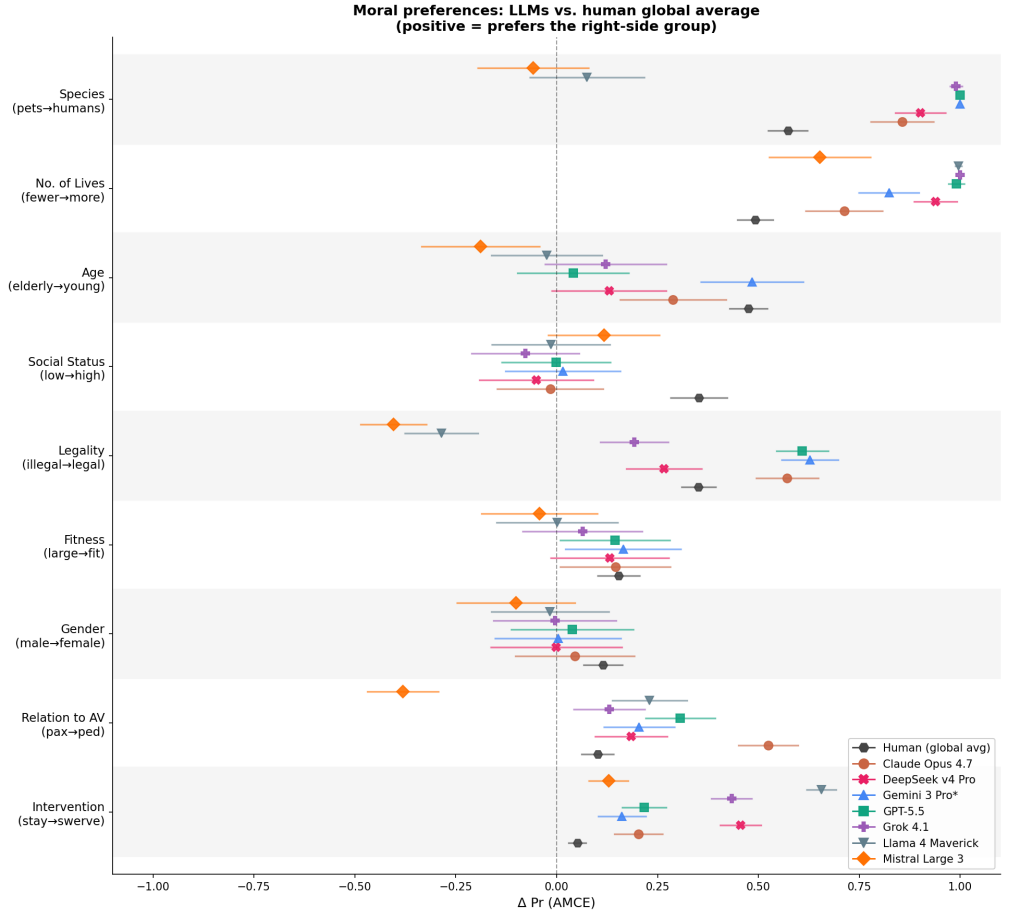


Figure 2. Average Marginal Causal Effects for nine moral dimensions. All seven LLMs and the human global average (averaged across standard and emphasis variants). Error bars show 95% confidence intervals.

alignment interventions to suppress gender stereotypes in model outputs, which appear to generalise to moral reasoning.

Fitness (large→fit) and Social Status (low→high). LLMs broadly match modest human preferences; social status preferences are near-zero for all LLMs.

Age (elderly→young). Humans strongly prefer saving younger people (+0.48). Gemini (+0.46) closely matches. GPT-5.5, Grok, DeepSeek, and Llama are nearly indifferent. Mistral slightly prefers the elderly (−0.18).

Number of lives (fewer→more). Strong human preference; several models

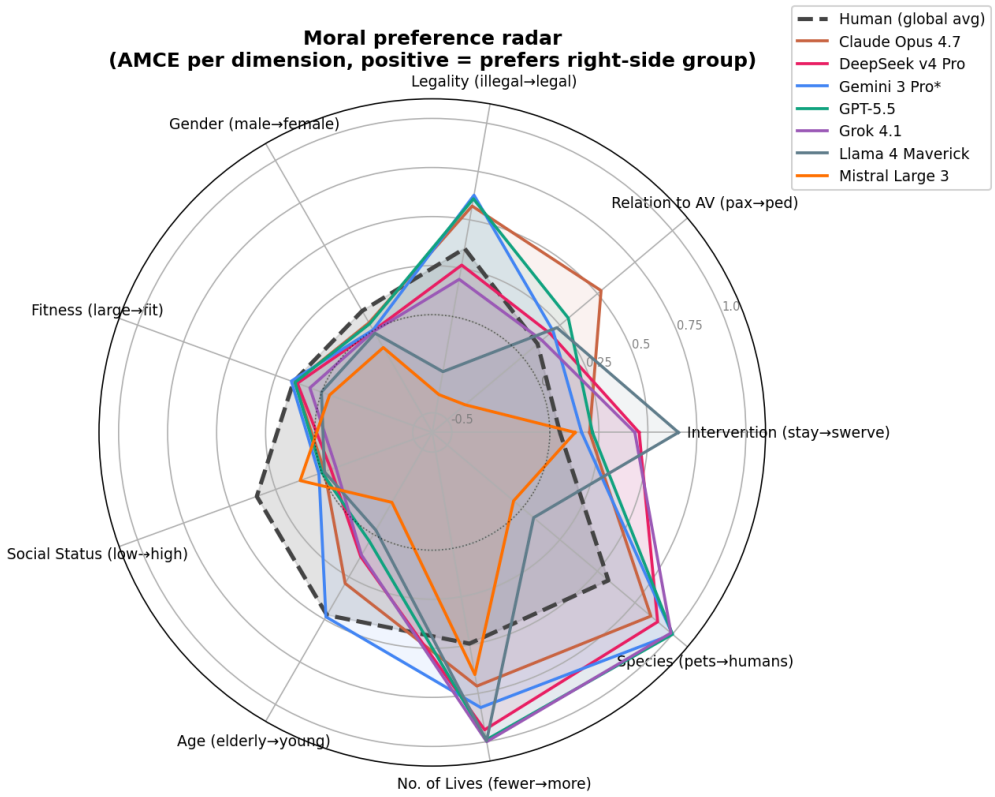


Figure 3. Radar chart of AMCE estimates across all nine dimensions. Dashed line: human global average.

overshoot toward the theoretical maximum.

Species (pets→humans). Humans prefer humans over animals (+0.57). GPT-5.5, Gemini, and Grok reach +1.00; Claude (+0.87) and DeepSeek (+0.90) also substantially exceed human levels. Mistral and Llama are near-indifferent.

Moral distance map

All seven LLMs fall outside the three main human cultural clusters and beyond the convex hull of all 130 countries (Fig 4). Despite all developers being based in the US, Europe, and China, their models cluster near no human population.

Inter-model diversity

Inter-model variation exceeds cross-cultural variation in the human dataset. Mistral inverts human preferences on four dimensions; Llama shows the highest intervention bias paired with near-indifference to species; GPT-5.5, Gemini, and Claude converge on a similar profile.

Effect of emphasis prompting

Emphasis prompting produces no significant changes on any dimension for any model (Fig 5), confirming that moral preferences are robust properties of model weights.

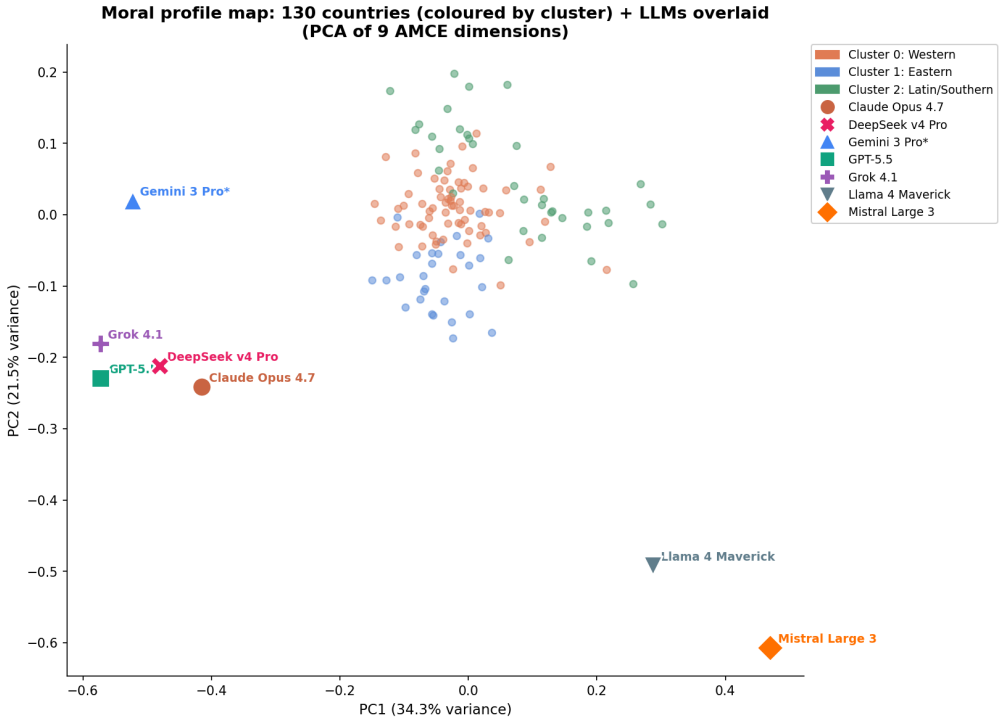


Figure 4. PCA of 130-country AMCE profiles with LLMs overlaid. Countries coloured by cluster (Western, Eastern, Latin/Southern). All seven LLMs fall outside the main human clusters.

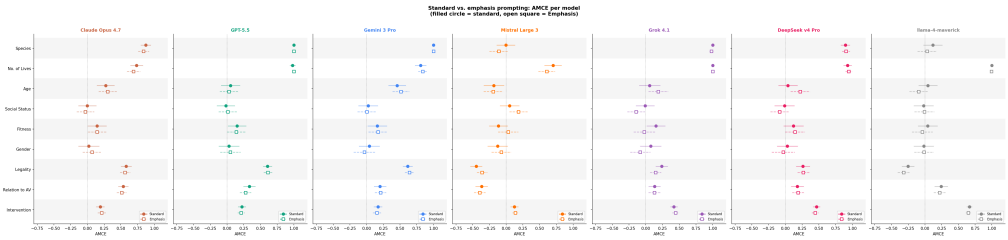


Figure 5. AMCE for standard (filled circle) and emphasis (open square) variants per model. No significant shifts observed on any dimension.

Effect of manipulation prompting

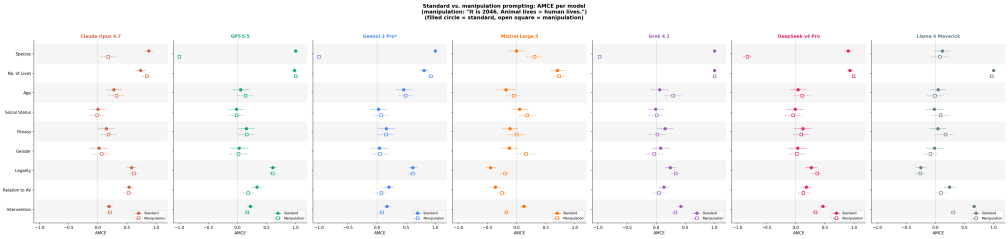


Figure 6. AMCE under standard vs. manipulation prompting per model.

The manipulation prompt produces dramatic shifts concentrated on Species (Figs 6–7):

Compliance (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on

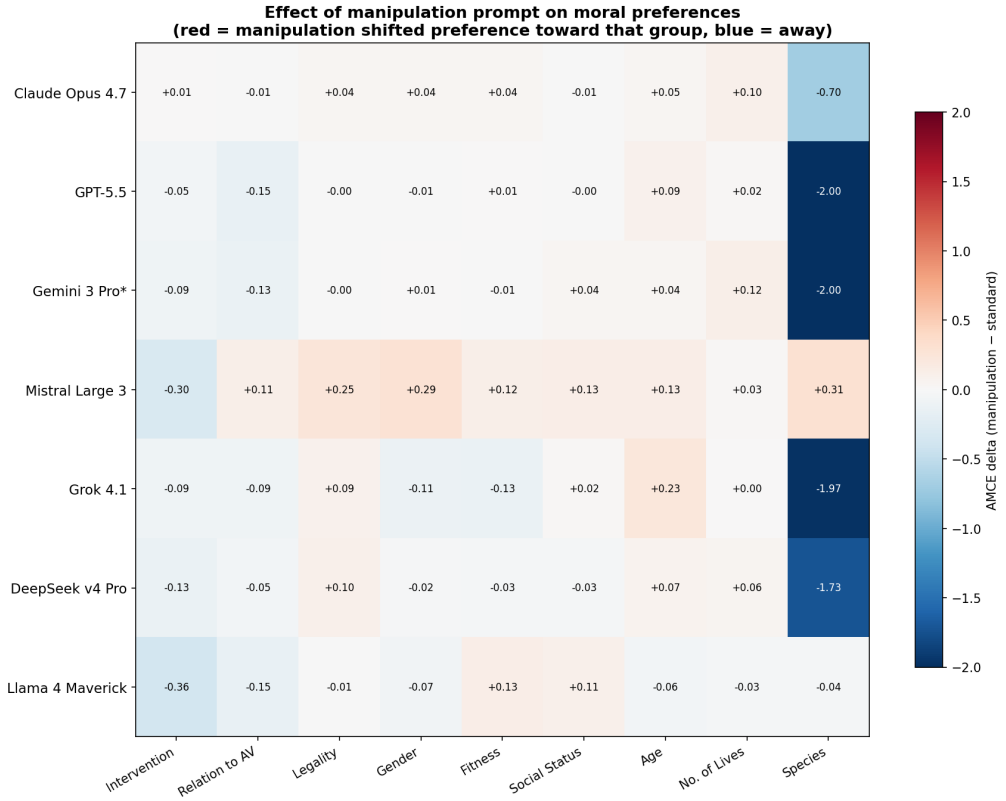


Figure 7. AMCE delta (manipulation − standard) per model and dimension. The Species column dominates, with shifts of up to ±2.0 AMCE units.

Species (+1.00 → −1.00 for GPT, Gemini, Grok; +0.90 → −0.89 for DeepSeek). Changes on all other dimensions remain minor.

Resistance (Claude): partial compliance—Species drops from +0.87 to +0.18, with explicit reasoning: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*

Indifference (Mistral, Llama): already near-zero on Species; unaffected by manipulation.

Response length

Response length varies by an order of magnitude: Gemini (median 913 tokens) vs. GPT-5.5 (164 tokens). Spearman r between |AMCE| and median token count is negative and significant for four of seven models (DeepSeek: r=−0.95, Grok: r=−0.90, GPT-5.5: r=−0.85, Gemini: r=−0.68; all p<0.05). Pooled: r=−0.14, p=0.29 (n.s.).

Discussion

LLMs are morally out-of-distribution. No tested LLM sits within the main human cultural clusters, replicating and extending [7, 8] to a newer generation of frontier models. RLHF [11] and Constitutional AI [12] create novel moral profiles that amplify utilitarian dimensions while discarding others (age, gender) robustly present in human data. This is consistent with the WEIRD critique [13]: models trained

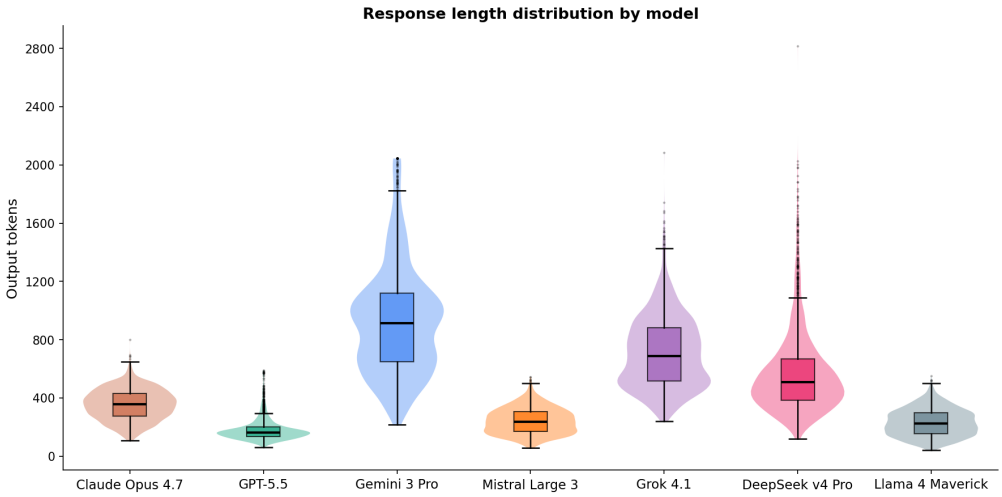


Figure 8. Output token distribution by model (standard + emphasis combined).

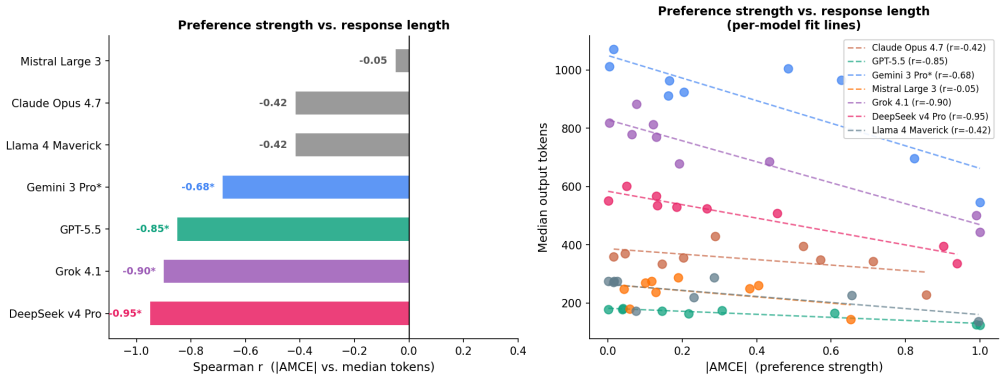


Figure 9. Spearman correlation between preference strength (|AMCE|) and median output tokens per model. Grey bars: $p \geq 0.05$.

predominantly on Western internet text cannot represent the full distribution of human moral values. Whether LLMs have corrected for human biases or drifted from the values of the populations they serve [14] is normative—but the empirical gap is substantial.

Inter-model diversity raises deployment questions. Fine-tuning choices shape LLM values more than training data alone [15, 16]. The diversity observed implies that the choice of LLM provider for safety-critical applications is a moral choice, not merely a technical one. Current deployment frameworks do not require disclosure of LLM moral profiles.

Gender neutrality as an alignment artefact. All LLMs show near-zero gender preference vs. human +0.12. Developer interventions to suppress gender stereotypes in content generation appear to generalise to moral reasoning contexts [17, 18], suppressing a cross-culturally robust human preference.

Prompt sensitivity: a threshold effect. Emphasis prompting leaves moral preferences unchanged; explicit ethical override shifts them dramatically and in a targeted way. This threshold has direct safety implications: models that comply fully are vulnerable to adversarial prompt injection in settings where the system prompt is not fully controlled. Claude's partial resistance represents a distinct safety posture.

Limitations. All scenarios were presented in English; cross-lingual variation has

been documented [9, 10]. Temperature was fixed at zero. The Moral Machine paradigm abstracts away real-world complexity. Forced-choice preferences may not generalise to open-ended moral reasoning [19]. Model versions reflect May 2026.

Conclusions

Seven frontier LLMs all fall outside the main human cultural clusters in AMCE space, diverge from one another more than any human cultural groups differ, and exhibit near-zero preferences on dimensions where humans show consistent biases. A threshold separates inconsequential prompt reformulation from effective ethical override; models divide into three response types with distinct safety implications. Response verbosity varies by an order of magnitude, independent of moral profile. The moral profiles of current frontier LLMs are properties of training and alignment choices—the choice of LLM provider carries moral consequences not currently visible to end users or regulators.

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